NETFLIX AFTER HOURS
PATH C: EXECUTION BACKLOG & JIRA TICKETS

Document Version: 1.0-FINAL
Sprint Backlog Breakdown

1. EPIC ARCHITECTURE & DEPENDENCY MAPPING

This backlog converts the PRD, Path A Technical Specs, and Path B Experimentation Strategy into actionable Jira tickets structured across 4 core Epics.

> **Epic 1: Core Player Controls → Epic 2: Audio DSP & OLED Dimming → Epic 3: Biometric Telemetry → Epic 4: Decay Bookmarking & Feature Flags**

2. EPIC 1: CORE PLAYER UI CONTROLS & MANUAL SLEEP TIMER

NAH-101 Low-Luminescence Moon Icon & Timer Modal Overlay

Type: User Story | Priority: High | **5 SP**

User Story: As a late-night viewer, I want a low-luminescence sleep timer icon in the player control bar so that I can configure bedtime parameters without screen glare.

Acceptance Criteria (Gherkin):

```
Scenario: Tap Sleep Timer Icon in Player Bar Given the user is watching a video stream on Mobile, Tablet, or TV 65" When the user taps or selects the "Moon Icon" in the player control bar Then a low-luminescence modal overlay opens with background #000000 (80% opacity) And displays preset timer chips: [15m], [30m], [45m], [End of Episode], [Custom] And active focus rings use high-contrast circadian amber (#FF4500) with 56x56dp touch targets.
```

Technical Sub-Tasks:

- NAH-101-A: Add SVG Moon vector asset to design token library for iOS, Android, and WebOS.
- NAH-101-B: Build accessible modal overlay component enforcing 56x56dp touch target scale.
- NAH-101-C: Implement 3-second toast feedback state ("*Sleep Timer set for 30 minutes*").

3. EPIC 2: CLIENT-SIDE AUDIO DSP & OLED DIMMING ENGINE

NAH-201 10-Minute Exponential OLED Display Dimming Curve

Type: User Story | Priority: High | **8 SP**

User Story: As a sleeping viewer, I want the screen brightness to decay exponentially to true black before timer expiry so that blue-light emissions do not disturb sleep.

```
Scenario: Exponential Screen Dimming Curve Given a sleep timer reaches T-10m prior to expiry When the dimming engine initiates Then luminance decreases along curve: L(t) = L_initial * (1 - t/600)^2 down to #000000 at T-0m And touching screen or remote button cancels dimming immediately and restores 100% luminance.
```

Sub-Tasks: NAH-201-A Platform luminance controllers (iOS/Android/TV) | NAH-201-B Touch/key listeners.

NAH-202 Client-Side 5-Minute Low-Pass Audio Filter & Volume Decay

Type: Tech Story | Priority: High | **13 SP**

User Story: As a viewer drifting off, I want audio frequencies and volume to step down gradually so that abrupt spikes do not startle me awake.

```
Scenario: Audio Frequency & Volume Attenuation Given timer reaches T-5m prior to expiry When DSP pipeline executes Then 2nd-order Butterworth filter ramps cutoff from 20kHz down to 1.8kHz and gain attenuates to -48dBFS.
```

Sub-Tasks: NAH-202-A Web Audio / C++ DSP BiquadFilter node | NAH-202-B DRC compressor tuning.

4. EPIC 3: BIOMETRIC TELEMETRY & WEARABLE SYNC

NAH-301 Apple HealthKit, Health Connect & Oura Integration — Type: User Story | Priority: Medium | **13 SP**

User Story: As a Premium subscriber wearing a smartwatch, I want Netflix to detect sleep onset so that bedtime mode triggers automatically.

```
Scenario: Wearable Sleep Onset Trigger Given a Premium user with linked Apple Watch, Health Connect, or Oura Ring
When wearable detects stage N1/N2 for >= 45s, client edge SDK receives sleep token And initiates decay sequence
while displaying a non-blocking 15-second TV overlay cancel prompt.
```

Sub-Tasks: NAH-301-A OS HealthKit / Health Connect listeners | NAH-301-B Oura webhook polling | NAH-301-C Zero-raw-data edge privacy.

5. EPIC 4: DECAY BOOKMARKING, RECOVERY CARD & FEATURE FLAGS

NAH-401 Offline-First Local Decay Bookmarking & Recovery Card — Type: User Story | Priority: High | **8 SP**

Acceptance Criteria: Save local timestamp at T-10m dim start into SQLite/Room; render morning Home feed hero card ("*Faded out at 14:20*") with 1-click resume.

NAH-402 LaunchDarkly Flag Routing & Automated Circuit Breaker — Type: Tech Story | Priority: Critical | **5 SP**

Acceptance Criteria: Route capabilities via LD rules; trip circuit breaker to Control within 30s if cancellation > 3.5% or error rate > 0.1%.

6. ALTERNATIVE PATHS & EDGE CASE MATRIX

Edge Case Scenario	System Reaction & Fallback	User Impact & Safeguard
Multi-Viewer Room	Wearable detects sleep for User A, but User B presses remote button.	Remote keypress immediately cancels sleep sequence and resets timer.
Mid-Fade Ad Break	Ad scheduled during active dimming curve (Standard with Ads).	Ad insertion engine pauses ad queue until dimming resolves or completes.
Wearable Disconnect	Bluetooth disconnects mid-stream on Apple Watch.	Client falls back seamlessly to scheduled timer without throwing UI errors.
Network Loss Overnight	Wi-Fi disconnects after local decay bookmark write.	Local SQLite queue buffers write; morning card renders offline from DB.

7. 4-SPRINT RELEASE ROADMAP ALLOCATION SCHEDULE

Sprint	Core Focus	Tickets Included	Milestone Deliverable
Sprint 1	Foundation & Player UI	NAH-101, NAH-402	Working timer modal & LaunchDarkly flags.
Sprint 2	Audio DSP & Display Dimming	NAH-201, NAH-202	Functional 10m dimming & 5m audio decay.
Sprint 3	Offline Caching & Recovery	NAH-401	Local SQLite storage & Home recovery card.
Sprint 4	Wearables & Experimentation	NAH-301	Apple/Oura sync, 5% A/B rollout, circuit breaker.